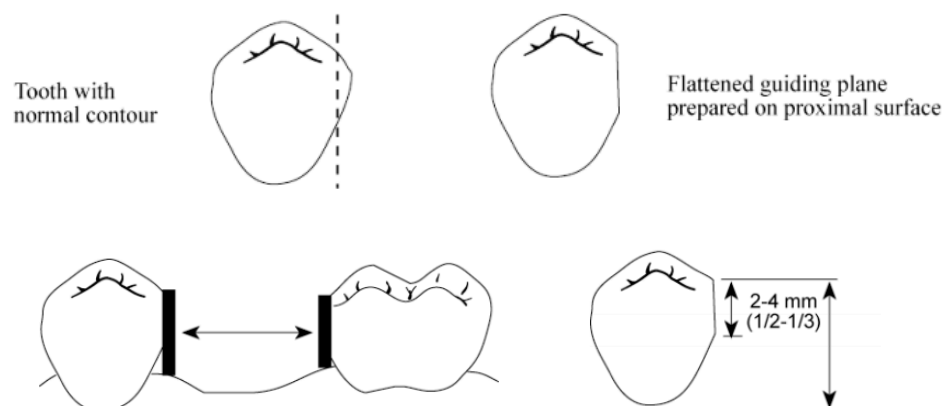


Surveying (Part 2)

Learning objectives:

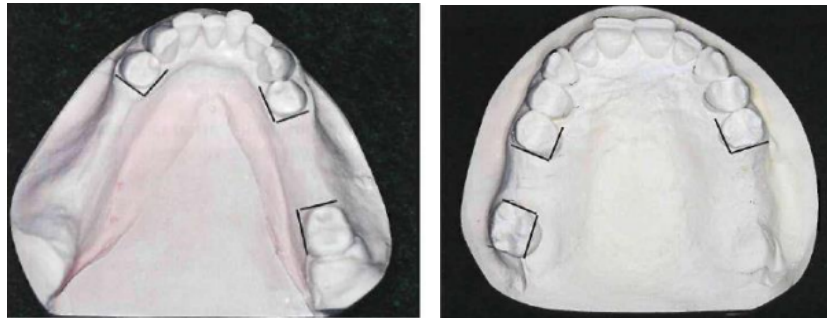
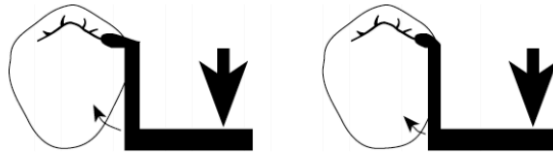
- Define guiding planes in RPD and their functions.
- Distinguish different types of undercuts.
- Describe the path of insertion of RPD

Guiding planes are two or more vertically parallel surfaces of abutment teeth shaped to direct the prosthesis during placement and removal. They may occur naturally on teeth, but more commonly need to be prepared. They are found or formed in the proximal or lingual surfaces of the abutment teeth in the occlusal third (generally a minimum of 2 - 4 mm in height). Guide planes are delineated by the surveyor.



Functions of the guide plane:

1. Guide the prosthesis in and out of the mouth without any undesirable forces against remaining teeth (Ensure a definite path of insertion).
2. The frictional forces of contact of the prosthesis with the guiding plane wall will contribute significantly to the retention of removable partial dentures.
3. It can provide bracing or stabilisation.



Types of undercuts established by the surveyor:

1. **Desirable undercuts:** are used for retaining the removable partial denture against the dislodging forces by incorporating a retentive flexible clasp arm.

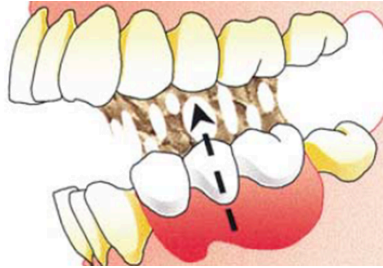


2. **Undesirable undercuts:** undercuts other than those used for retention are considered undesirable and should be eliminated by:
 - a. Tooth re-contouring.
 - b. Placing a properly contoured crown restoration on the tooth.
 - c. Tilting the cast and changing the path of insertion.
 - d. Blocking out the undercut with wax on the master cast



A path of insertion/removal is the path along which a prosthesis is placed (or removed) intraorally. A removable partial denture is usually fabricated to have a single path of insertion or removal (withdrawal) from the mouth.

This path may or may not coincide with the path of displacement. There may be a single path or multiple paths of insertion.



***Path of displacement:** This is the direction in which the denture tends to be displaced in function. The path is variable, but is assumed for design to be at right angles to the occlusal plane.

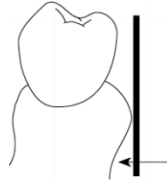
Factors that affect the path of insertion and removal

1. Guiding planes.
2. Retentive areas: retentive areas must exist for a given path of placement and must be contacted by retentive clasp arms that are forced to flex over a convex surface during placement and removal. For a clasp to be retentive; its path of escapement must be other than parallel to the path of removal of the denture itself.

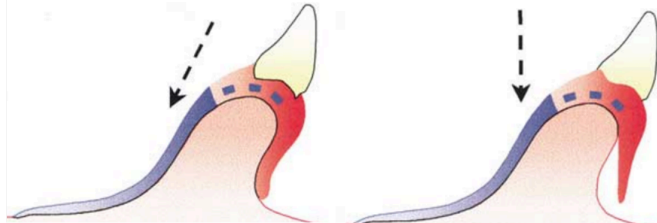
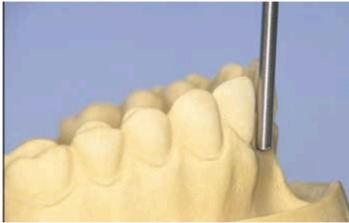


In the graph, the clasp has a different path of escapement (dashed arrow) than the guiding plane (solid arrows) and therefore must flex when the denture is removed. This provides retention for the denture.

3. Interference: the prosthesis must be designed so that it may be placed and removed without encountering tooth or soft tissue interference for example severely inclined teeth, bone exostosis, and tissue undercuts.



4. Esthetics: by one path of placement, the most esthetic location of artificial teeth is made possible, and less clasp metal and base material may be displayed.



Recording the cast tilt:

If the cast has been tilted for the final survey, the degree of tilt must be recorded so that the position of the cast can be reproduced in the laboratory. There are two methods of recording the degree of tilt.

1. Using the tripod method, the vertical arm of the surveyor is locked at a height that allows the tip of the marker to contact the palatal surface of the ridge in the molar and incisal regions. Three points are marked with the graphite marker, one on each side posteriorly and one anteriorly. The points will then be ringed with a pencil so that they are clearly visible. (Figure 1)
2. Alternatively, the analysing rod is placed against one side of the base of the cast, and a line is drawn on the cast parallel to the rod. This is repeated on the other side and at the back of the cast so that there are three widely spaced lines parallel to the path of insertion. (Figure 2)

Figure 1

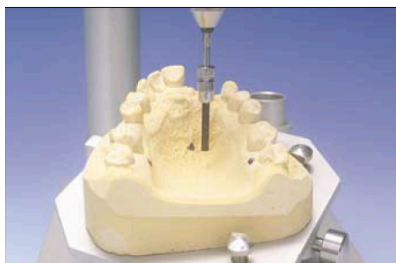


Figure 2

